HelioX RAG 2.1 — Technical Architecture Blueprint

1. System Overview

HelioX RAG 2.1 is a Retrieval-Guided Multi-Agent Deliberative Architecture designed for: - Low-latency selective retrieval - Reduced token consumption - High-grounded reasoning - Structured hallucination mitigation

Core Principles: 1. Deterministic preprocessing before probabilistic reasoning. 2 Metadata-first retrieval, embedding-second retrieval. 3. Selective expert activation instead of full-context reasoning. 4. Evidence-bound generation (no uncited claims). 5. Confidence-aware output.

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2. Macro Architecture

User Query ↓ Query Analyzer (Intent + Constraint Extractor) ↓ Routing Engine (Expert Selector) Worker Agents (Chunk Reasoners) ↓ Adjudication Engine (Conflict + Authority Resolver) ↓ Answ Composer (Citation-Bound Output) ↓ Confidence Scorer + Session Memory

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3. Phase 1 — Deterministic Sanitization Layer

3.1 Document Ingestion Pipeline

Input Types: - PDF (native + scanned) - CSV - TXT - HTML - OCR documents

Pipeline Steps: 1. Format Detection 2. Structural Parsing - Section detection - Table detection Footnote mapping 3. Noise Removal - Header/footer stripping - Page number removal - Encod normalization 4. Parent–Child Graph Construction - Section → Subsection → Paragraph → Sentenc Table → Row → Cell mapping

Output: Structured Document Graph (JSON)

3.2 Ontology & Terminology Standardization

Instead of rewriting content via LLM: - Domain dictionary mapping - Synonym expansion inde Abbreviation resolution table

Output: Normalized semantic tokens without altering original meaning.

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4. Phase 2 — Multi-Vector Expert Profiling Layer

Each 800–1200 token chunk produces:

1. Primary Semantic Embedding
2. Entity Vector (NER extracted entities)
3. Hypothetical Question Embeddings (HyDE generation)
4. Constraint Metadata:
5. Temporal
6. Geographic
7. Regulatory
8. Applicability scope

Storage Architecture: - Vector DB (FAISS / Qdrant / Weaviate) - Metadata DB (PostgreSQL / Ela
Cache (Redis)

Expert Profile Schema: { chunk_id, semantic_embedding, entity_list, hyde_questions, constraint_tags, authority_score, timestamp, parent_section }

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5. Phase 3 — Intent-Driven Routing Engine

5.1 Query Decomposition

Input Query → Structured Query Object

{ fact_components, reasoning_components, constraint_filters, entity_targets }

Steps: 1. Intent classification (fact lookup / comparison / synthesis / compliance check) .2 En
extraction 3. Temporal & jurisdiction filter extraction

5.2 Hybrid Retrieval Strategy

Stage 1: Metadata Filtering Stage 2: Entity Matching Stage 3: Vector Similarity Search Stage
Fallback Broad Retrieval (if recall < threshold)

Top-K Selection: Dynamic K based on confidence score.

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6. Phase 4 — Parallel Worker Agents

Each worker receives: - One chunk - Structured query - Mandatory citation format

Worker Output Schema: { claim, supporting_text_span, reasoning_trace, confidence_local }

Rules: - No claim without span index - No external knowledge allowed

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7. Phase 5 — Adjudication Engine

7.1 Conflict Detection

- Semantic contradiction detection
- Overlap inconsistency scoring

If conflict detected: → Trigger re-evaluation with stricter extraction prompt

7.2 Authority Weighting

Weighted score = (w 1 × recency) + (w 2 × source_rank) + (w 3 × citation_density) + (w worker_confidence)

Highest weighted consensus wins.

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8. Phase 6 — Evidence-Bound Answer Composer

Rules: 1. Every factual sentence must map to citation. .2 If coverage < threshold → out "Insufficient evidence." 3. Structured output template enforced.

Output Format: - Direct Answer - Supporting Evidence (citations) - Constraints Applied - Confidence Score

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9. Phase 7 — Confidence & Self-Validation Layer

Metrics: 1. Retrieval coverage score 2. Citation density 3 Conflict severity score 4 Cross-worker agreement score

Global Confidence = weighted aggregation.

If confidence < X: → Trigger expanded retrieval OR request clarification from user.

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10. Session Memory (Controlled)

Memory Stores Only: - Validated structured summary - Confirmed constraints - User preference context

Memory Never Stores: - Raw LLM speculation - Unverified claims

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11. Hallucination Mitigation Strategy

1. Deterministic preprocessing
2. Retrieval-first reasoning
3. Mandatory citation enforcement
4. Cross-agent contradiction detection
5. Confidence threshold gating
6. Fallback retrieval safety net

Expected Outcome: Hallucination significantly reduced but not mathematically zero.

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12. Scalability Considerations

- Async parallel worker execution
- Embedding caching
- Batch ingestion pipeline
- Horizontal scaling of worker layer
- Vector sharding for large corpora

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13. Evaluation Framework

Offline Metrics: - Retrieval Recall@K - MRR - Citation Accuracy - Faithfulness Score

Online Metrics: - Latency per query - Token consumption - Conflict trigger frequency - Confic calibration curve

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14. Architectural Improvements Over 2.0

1. Hybrid deterministic + semantic routing
2. Fallback recall safety layer
3. Confidence-aware gating
4. Authority-weighted adjudication
5. Strict evidence-bound generation
6. Structured worker schema

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15. End-to-End Workflow Diagrams

Below are clear execution flows for: 1. Data Upload (Ingestion Path) 2. User Query (Inference Path)

A. DATA UPLOAD WORKFLOW (PDF / CSV / TXT / OCR)

[User Uploads File] ↓ [File Type Detection] ↓ [Parser Layer] - PDF parser / OCR engine - C reader - HTML cleaner ↓ [Noise Removal Engine] - Header/footer stripping - Page number rer Encoding normalization ↓ [Structural Decomposition] - Section detection - Table extraction - Parent- graph creation ↓ [Semantic Standardization] - Abbreviation resolution - Ontology mapping - Synor expansion ↓ [Chunking Engine] - 800–1200 token adaptive chunks - Maintain parent sectic reference ↓ [Expert Profiling Generator] For each chunk: • Semantic embedding • Entity extraction HyDE question generation • Constraint tagging (time/geo/scope) ↓ [Storage Layer] - Vector DB (ser HyDE vectors) - Metadata DB (entities + constraints) - Redis (hot cache index) ↓ [Blackboard Updated] ↓ System Ready for Query

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B. USER QUERY WORKFLOW (INFERENCE PATH)

[User Submits Query] ↓ [Query Analyzer] - Intent classification - Entity extraction - Constraint ex Fact vs Reasoning separation ↓ [Structured Query Object Created] ↓ [Routing Engine – Hybrid Stage 1: Metadata filtering (constraints) Stage 2: Entity matching Stage 3: Vector similarity sear Stage 4: Confidence check

If recall < threshold: → Trigger fallback broad retrieval ↓ [Top-K Expert Chunks Selected] ↓ [Pa Agents Activated] Each worker receives: • One chunk • Structured query • Citation rules

Worker Output: • Claim • Supporting span • Local confidence ↓ [Adjudication Engine] - De contradictions - Authority weighting - Agreement scoring

If conflict detected: → Re-evaluation prompt ↓ [Answer Composer] - Evidence-bound generation Structured output format - Inline citations ↓ [Confidence Scoring Layer] - Retrieval coverage - density - Worker agreement score

If confidence < X: → Expand retrieval OR ask user clarification ↓ [Final Answer Delivered] ↓ Memory Update] - Store validated structured summary only

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C. SYSTEM CONTROL LOGIC (SIMPLIFIED FLOW)

Upload → Sanitize → Profile → Index → Ready Query → Analyze → Route → Parallel Reason Validate → Respond

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Final Statement

HelioX RAG 2.1 operates as two major pipelines: 1. A deterministic ingestion pipeline that bu structured expertise. 2. A selective multi-agent inference pipeline that activates only relevant knowledge.

This separation is what enables speed, cost efficiency, and strong hallucination mitigation.